
Professional Summary

Ph.D. researcher in quantum computing and quantum networking, with graduate training in theoretical physics, quantum information, quantum optics, and open quantum systems. Research spans quantum network protocol design, entanglement and nonlocality certification, quantum channel learning, Pauli-noise characterization, and quantum-circuit verification, supported by benchmark-driven numerical experiments. Builds Python/C++ pipelines for simulation, validation, and reproducible data aggregation, with hands-on IBM Quantum cloud-device experience.

Education

09/2023 – Ph.D. in Computer Science

Present *Stony Brook University, New York, United States*

Advisor: Nengkun Yu

09/2022 – Master of Science in Quantum Information

08/2023 *Sorbonne Université, Paris, France*

Thesis: “NONLOCALITY DISTILLATION”

Relevant training: Quantum information, quantum optics, photonic quantum computing

09/2021 – Master of Science in Theoretical Physics

08/2022 *École normale supérieure (ENS) de Paris, France*

Thesis: “CLASSICAL SIMULATION OF SHALLOW RANDOM QUANTUM CIRCUITS”

Relevant training: Quantum field theory, statistical physics, many-body physics

09/2018 – Engineering Degree (Ingénieur polytechnicien)

08/2021 *École Polytechnique, Institut Polytechnique de Paris, France*

Specialization: Fundamental Physics

Research Experience

09/2023 – Research Assistant

Present *Department of Computer Science, Stony Brook University*

Advisor: Nengkun Yu

- Contribute to quantum network design research, including a verification-based quantum transmission protocol and nonlocality-based entanglement certification with applications to quantum networks.
- Study quantum channel learning and Pauli-noise characterization, developing noise-estimation and benchmarking methods relevant to near-term and fault-tolerant quantum computing.
- Design and implement Python/C++ software for scalable quantum circuit equivalence checking and verification (symbolic methods, local-projection fingerprints, weighted binary decision diagrams, simulation-based validation), with automated benchmark suites over structured and randomized Clifford+ T circuits for reproducible empirical evaluation.
- Run and analyze quantum-circuit experiments on IBM Quantum cloud devices, comparing hardware data with simulator expectations and circuit-level noise models drawn from open-system dynamics.

05/2023 – Research Internship

09/2023 *Center for Theoretical Physics (IPhT), Commissariat à l'Énergie Atomique (CEA), France*

Project: “NONLOCALITY DISTILLATION”

- Developed optimization-based methods for quantum nonlocality distillation with applications to quantum communication and quantum key distribution.
- Formulated quantum-information constraints as linear programs and analyzed numerical outputs under different protocol assumptions.

04/2022 – Research Internship

07/2022 *Inria de Lyon, École normale supérieure (ENS) de Lyon, France*

Project: “CLASSICAL SIMULATION OF SHALLOW RANDOM QUANTUM CIRCUITS”

- Developed semidefinite-programming-based methods for classical simulation of shallow random quantum circuits.
- Translated multi-qubit circuit models into computational optimization problems and evaluated simulation behavior across circuit families.

03/2021 – Research Internship

07/2021 *Irène Joliot-Curie Laboratory of Physics of Two Infinities (IJCLab), Orsay, France*

Project: “PDF UNCERTAINTIES AND THE W BOSON MASS MEASUREMENT”

Analyzed the contribution of parton distribution function uncertainties to the systematic uncertainty of the W boson mass measurement in the ATLAS experiment.

Selected Publications

- 06/2026 **Rethinking Quantum Network Design Using a Verification-Based Quantum Transmission Protocol**, Yiming Zeng, Zhengyu Wu, **Xuan Du Trinh**, Yuanyuan Yang, Nengkun Yu, Aruna Balasubramanian, *46th IEEE International Conference on Distributed Computing Systems (ICDCS 2026)*; **Distinguished Paper Award**
- 02/2026 **On the Emergence of Classical Stochasticity**, **Xuan Du Trinh**, Ismaël Septembre, Hai-Chau Nguyen, [arXiv:2602.04633 \[quant-ph\]](#)
- 01/2026 **Manjushri: A Tool for Equivalence Checking of Quantum Circuits**, **Xuan Du Trinh**, Meghana Sistla, Nengkun Yu, Thomas Reps, [arXiv:2601.22372 \[quant-ph\]](#)
- 10/2025 **Scalable Equivalence Checking and Verification of Shallow Quantum Circuits**, Nengkun Yu, **Xuan Du Trinh**, Thomas Reps, [Proceedings of the ACM on Programming Languages, Vol. 9, OOPSLA2](#)
- 09/2025 **Adaptivity Is Not Helpful for Pauli Channel Learning**, **Xuan Du Trinh**, Nengkun Yu, [Quantum, 9, 1864](#)
- 07/2025 **Entanglement Certification by Measuring Nonlocality**, **Xuan Du Trinh**, Zhengyu Wu, Junlin Bai, Huan-Hsin Tseng, Nengkun Yu, Aruna Balasubramanian, [arXiv:2507.18066 \[quant-ph\]](#)

Skills

Programming	Python, C/C++, Julia, MATLAB, Java
Quantum networking	Quantum network protocol design, verification-based quantum transmission, entanglement and nonlocality certification
Quantum computing	Quantum channel learning, Pauli-noise characterization, tool benchmarking, quantum circuit simulation and verification, Qiskit, Quasimodo
Physics	Quantum information, quantum optics, open quantum systems, photonic quantum computing, quantum field theory, statistical physics, many-body physics
Research tools	Linux, Git/GitHub, reproducible workflows, Mathematica, ROOT
Mathematics	Probability and statistics, numerical linear algebra, linear programming, semidefinite programming, numerical optimization, algorithms
Languages	Vietnamese (Native), English (Proficient), French (Proficient)

Teaching, Mentoring, Service, and Honors

- Teaching Teaching Assistant: Foundations of Computer Science (2023); Analysis of Algorithms (2025).
- Mentoring Research Mentor (Summer 2025): mentored a high school student selected as a **Top 300 Scholar** in the **85th Regeneron Science Talent Search**.
- Service Reviewer: IEEE ICDCS 2024; IJCNN 2024 Quantum Machine Learning Track.
- Honors Distinguished Paper Award (ICDCS 2026); Chairman Fellowship (Stony Brook CS); VALLET Scholarship (Vallet Foundation); EIFFEL Scholarship (French Government)